GHraisoni COLLEGE
Engineering and Management
Nagpur

RSOC-2026

RSOC 2026

GROUP 28

Domain :– Data Science

Problem Statement Title:- AI-Powered Urban Insights Platform

Team Leader Name: - Prakash Gond

Group No:- 28

Team Name:- Threads

Members:-

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- Vaishnavi Nandurkar

NagarMitra

- AI-powered platform that integrates traffic, pollution, energy, transport, water, and waste data into one smart city analytics system.
- Uses data pipelines, machine learning, and interactive dashboards to monitor and manage urban systems.
- Provides real-time insights, predictive analytics, and decision-support tools for city authorities.

How It Addresses the Problem

- **Unified Data Platform:** Aggregates fragmented urban datasets into a single system.
- **Predictive Analytics:** ML models forecast traffic congestion, pollution levels, and transport demand.
- **Anomaly Detection:** Automatically detects pollution spikes, traffic surges, and energy overconsumption.
- **Policy Simulation:** Allows planners to test interventions (e.g., reduce traffic by 20%) and evaluate their impact.
- **Interactive Dashboard:** Real-time maps, alerts, and analytics for faster decision-making.

Innovation & Uniqueness

- Integrated Smart City Digital Twin combining multiple urban domains.
- AI-driven decision support (predictions + alerts + simulations).
- Ai-chatbot using vector-less RAG.
- Modular architecture enabling scalable and rapid development.

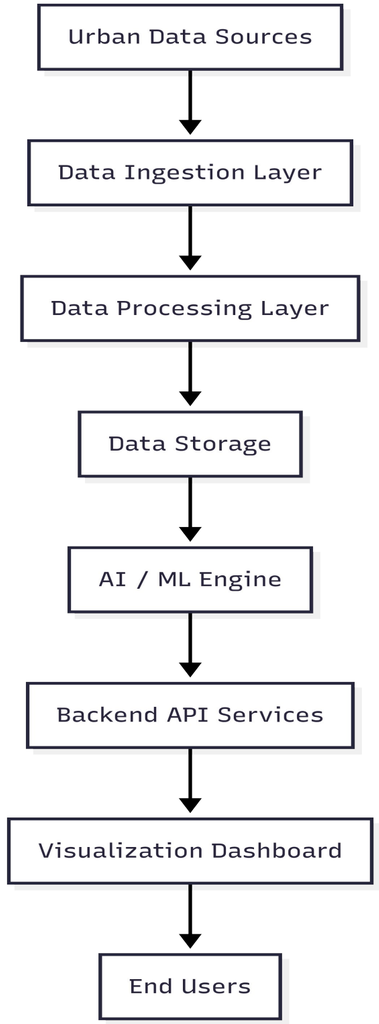
FEASIBILITY AND VIABILITY

- **Feasibility of the Idea**
  - Uses widely available technologies (Python, FastAPI, ML models, Streamlit, Firebase).
  - Can operate with synthetic or open urban datasets for prototyping.
  - Modular architecture allows parallel development and easy integration.
  - Scalable design suitable for future real-world smart city deployments.
- **Potential Challenges & Risks**
  - Limited availability of real-time city data from sensors and authorities.
  - Data quality issues (missing, noisy, or inconsistent datasets).
  - Model accuracy limitations due to synthetic or limited training data.
  - System integration complexity across data pipeline, ML models, and dashboard.
- **Strategies to Overcome Challenges**
  - Use public datasets and simulated IoT streams during development.
  - Implement data cleaning, validation, and normalization pipelines.
  - Continuously retrain models with real datasets as they become available.
  - Follow clear API contracts and modular architecture for smooth integration.

Impacts and Benefits & System Architecture

- **Potential Impact on Target Audience**
  - Enables city authorities and planners to make faster, data-driven decisions.
  - Helps transport and infrastructure departments monitor congestion and resource usage.
  - Improves public awareness and transparency through accessible city analytics.
  - Supports policy makers in evaluating urban interventions before implementation.

- **Benefits of the Solution**
  - **Social:** Improved urban mobility, better air quality, and enhanced quality of life.
  - **Economic:** Optimized resource usage and reduced costs from traffic congestion and energy waste.
  - **Environmental:** Lower pollution levels through smarter traffic and energy management.
  - **Operational:** Real-time alerts and predictive insights enable proactive city management.



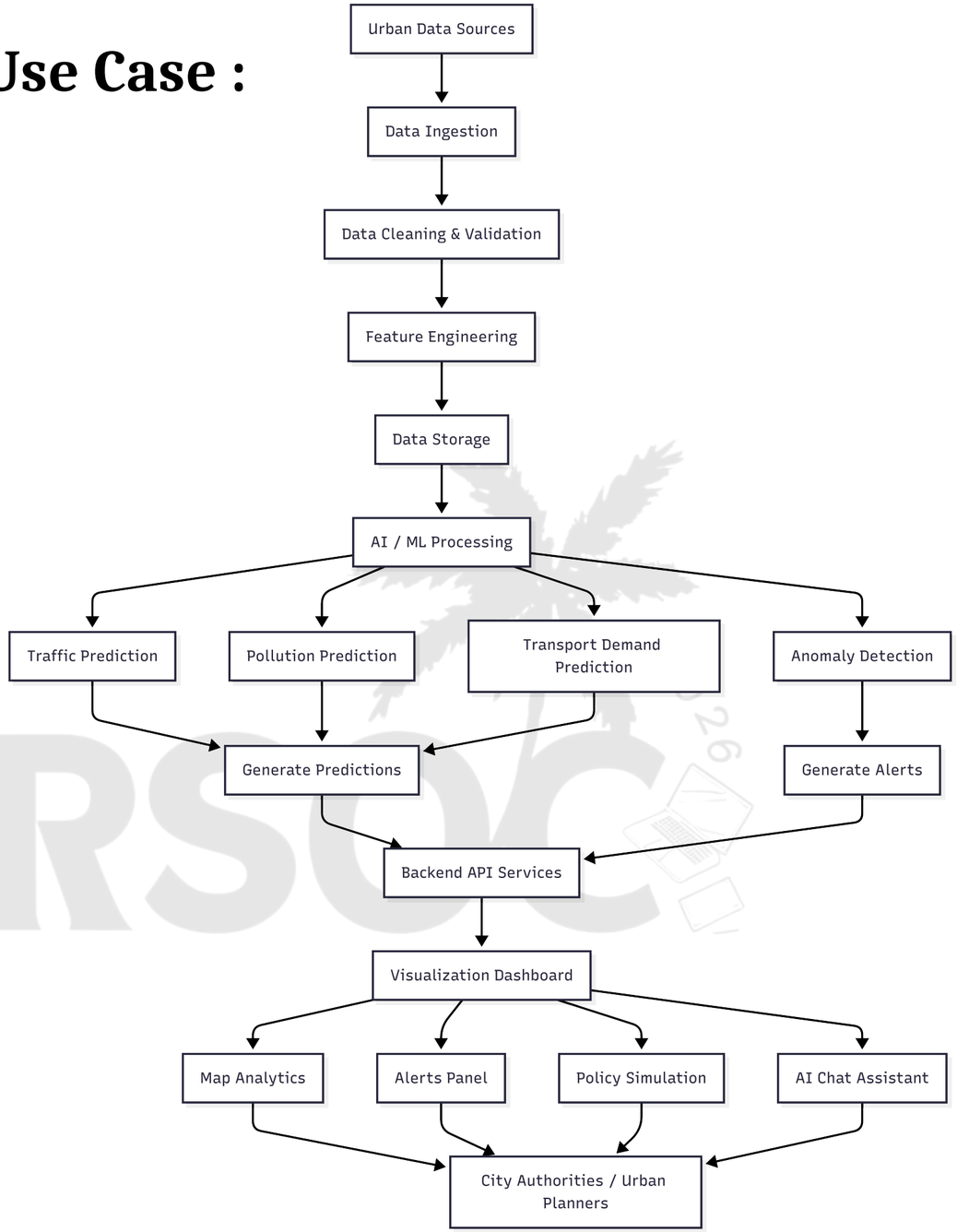
RESEARCH AND REFERENCES

- **Dataset :**
  - *dataset*

- **Research Paper :**
  - *Smart City Traffic Patterns Prediction Using Machine Learning*
  - *Machine Learning-Based Air Pollution Monitoring and Forecasting*
  - *Traffic Pattern Classification in Smart Cities Using Deep Recurrent Neural Networks*

- **News Articles :**
  - *Rising Vehicles Causing Severe Traffic Congestion*
  - *Bengaluru Among the Most Congested Cities in the World*
  - *Weather and Urban Conditions Increasing Air Pollution*
  - *Large Vehicles Increasing Urban Congestion and Safety Risks*

Use Case :


Urban Data Sources
Data Ingestion
Data Cleaning & Validation
Feature Engineering
Data Storage
AI / ML Processing
Traffic Prediction
Pollution Prediction
Transport Demand Prediction
Anomaly Detection
Generate Predictions
Generate Alerts
Backend API Services
Visualization Dashboard
Map Analytics
Alerts Panel
Policy Simulation
AI Chat Assistant
City Authorities / Urban Planners